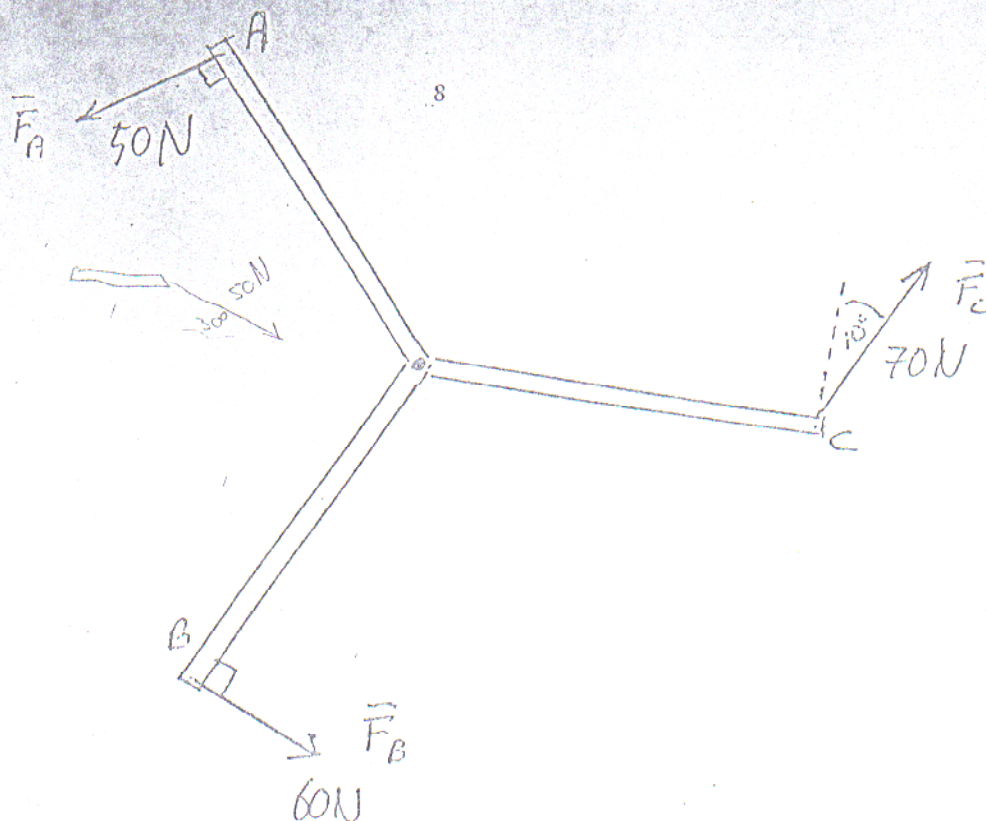




Toon aan dat die draaimoment is $\tau = 805,8 \text{ Nm}$. (Toon duidelik u bewerkings aan).
 Show that the torque is $\tau = 805,8 \text{ Nm}$. (Clearly show your calculations)

$$\begin{aligned}\tau_{\text{total}} &= \tau_1 + \tau_2 + \tau_3 = r_A F_A \sin \theta_A + r_B F_B \sin \theta_B + r_C F_C \sin \theta_C \\ &= 5,2 (50 \cos 30^\circ) \sin 90^\circ + 5,2 (60 \cos 30^\circ) \sin 90^\circ + 5,2 (70 \cos 30^\circ) \sin 80^\circ \\ &= 225,1666 + 270,1999 + 310,44 \\ &= 805,8 \text{ Nm}\end{aligned}$$

- (j) Wat is die hoekversnelling α ?
 Calculate the angular acceleration α .

$$\begin{aligned}\tau &= I \alpha \quad \therefore \alpha = \frac{\tau}{I} = \frac{805,8}{6,49 \times 10^{-3}} \\ &= 0,124 \text{ rad/s}^2\end{aligned}$$